Assignment 4: Linux Server Hardening

Objective: Secure a Linux server following industry best practices.

GitHub Repository Structure

assignment-4-linux-hardening/
■■■ README.md
■■■ screenshots/
■■■ configs/
■ ■■■ sshd_config
■ ■■■ ufw_rules.txt
■ ■■■ fail2ban_jail.local
■■■ troubleshooting.md
■■■ rollback-guide.md

Detailed README – Implementation Steps

1. Disable root login in /etc/ssh/sshd_config.
2. Configure SSH key authentication.
3. Disable password authentication.
4. Change SSH port from 22 to 2222.
5. Configure UFW firewall rules.
6. Install and configure Fail2Ban.
7. Remove unused packages and services.
8. Apply secure file permissions to sensitive files.

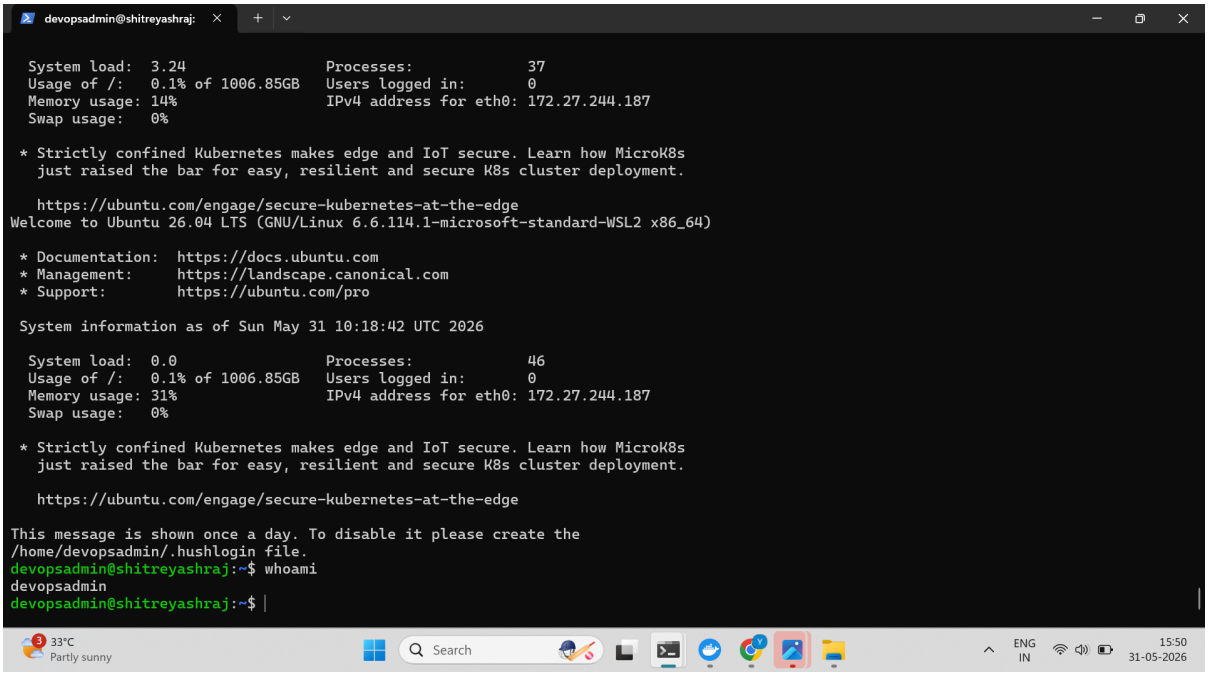
Hardening Checklist

✓ Root login disabled
✓ SSH key authentication enabled
✓ Password authentication disabled
✓ Custom SSH port configured
✓ UFW firewall enabled
✓ Fail2Ban installed and configured
✓ Unnecessary packages removed
✓ Secure permissions applied

Configuration Files Used

sshd_config: PermitRootLogin no PasswordAuthentication no Port 2222 UFW: ufw allow 2222/tcp ufw enable Fail2Ban: enabled = true port = 2222 maxretry = 5

Screenshot Evidence



Before and After Comparison

Before: - Root login allowed - Password authentication enabled - Default SSH port 22 - No firewall protection After: - Root login disabled - SSH key authentication enforced - Password authentication disabled - Custom SSH port configured - Firewall and intrusion protection enabled

Troubleshooting Guide

Issue: Unable to SSH after changing port.
Solution: Verify UFW rule and restart SSH service. Issue: Fail2Ban not banning IPs.
Solution: Check jail status and logs. Issue: Firewall blocks access.
Solution: Allow SSH port before enabling UFW.

Rollback / Recovery Procedure

1. Restore original sshd_config backup. 2. Re-enable password authentication if required. 3. Reset UFW rules. 4. Restart SSH service. 5. Verify access before logging out.

Key Decisions and Configurations

- SSH keys improve authentication security. - Custom SSH port reduces automated scans. - UFW provides host-level network protection. - Fail2Ban protects against brute-force attacks. - Principle of least privilege applied to file permissions.